GASTRAZOLE

(For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory)

Omeprazole For Injection 40 mg / Vial

Name and Strength of Active Ingredient
(Omeprazole Sodium BP equivalent to Omeprazole 40 mg)

Dosage form: Sterile Lyophilized cake for Injection

Product Description

Gastrazole (Omeprazole for injection 40 mg) is a white to almost white lyophilized powder/cake packed in USP type 1 amber glass lyo vial. The reconstituted solutions of Omeprazole for Injection were clear colorless solutions immediately after reconstitution and turned to slightly pale yellow color solutions at the end of 4 hours of storage at 30°C and complied with the acceptance criteria.

Pharmacodynamics

Omeprazole belongs to a class of antisecretory compounds, known as 'proton pump inhibitors' which are substituted benzimidazoles, that suppress gastric acid secretion by specific inhibition of the $H^+ K^+$ -ATPase enzyme system, the 'acid (proton) pump' of the gastric parietal cell. After intravenous administration of omeprazole, the onset of the antisecretory effect occurs within one hour, with the maximum effect occurring within two hours. A single dose of 40 mg of omeprazole given intravenously has similar effect on intragastric acidity over a 24-hour period as repeated oral dosing with 20 mg once daily. Although the plasma half-life of omeprazole is very short (~ 50 minutes), the antisecretory effect lasts longer due to prolonged binding to the parietal $H^+ K^+$ ATPase enzyme. When the drug is discontinued, secretory activity returns gradually, over 3 to 5 days. The inhibitory effect of omeprazole on acid secretion is dose-related and increases with repeated once daily dosing, reaching a plateau after four days.

During treatment with antisecretory medicinal products, serum gastrin increases in response to the decreased acid secretion. Also CgA increases due to decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumours.
Available published evidence suggests that proton pump inhibitors should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated following PPI treatment to return to reference range.

Pharmacokinetics
The apparent volume of distribution of omeprazole in healthy subjects and in patients with renal insufficiency is almost similar. The volume of distribution is slightly decreased in the elderly and in patients with hepatic insufficiency. The plasma protein binding of omeprazole is about 95%. The average half-life of the terminal phase of the plasma concentration time curve following intravenous administration of omeprazole is approximately 40 minutes. Omeprazole is completely metabolized by the cytochrome P450 system, mainly in the liver. No metabolite has been found to have any effect on gastric acid secretion. Almost 80% of an intravenous dose of omeprazole is excreted as metabolites in the urine, and the remainder is found in the faeces, primarily originating from biliary secretion.

Indication
Treatment of duodenal and gastric ulcer, NSAID associated gastric acid and duodenal ulcers or erosions, Helicobacter pylori eradication in peptic ulcer disease, reflux oesophagitis, symptomatic gastro-oesophageal reflux disease, acid related dyspepsia and Zollinger-Ellison syndrome.

Recommended Dose

Adults only

Prophylaxis of acid aspiration: Omeprazole 40 mg given as an intravenous infusion to be completed one hour before surgery.

Treatment in patients where oral therapy is inappropriate e.g. in severely ill patients with either reflux oesophagitis, duodenal ulcer or gastric ulcer: Omeprazole 40 mg given as an intravenous infusion once daily is recommended for up to 5 days.
The I.V. infusion produces an immediate decrease in intragastric acidity and a mean decrease over 24 hours of approximately 90%. Clinical experience in Zollinger-Ellison syndrome is limited.
Zollinger-Ellison syndrome: initially 60 mg IV daily, then adjust dosage individually. For doses exceeding 60 mg daily, give as divided doses twice daily.

Administration
Administration by intravenous route only and not by any other route Injection
For I.V. injection, each single dose vial containing lyophilized powder of Omeprazole 40 mg should be reconstituted with 10 ml of water for injection. The resulting concentration is 4 mg/ml of omeprazole. Omeprazole, 40 mg, should be given slowly (over a period of 5 minutes) as an intravenous injection.
Omeprazole powder for solution for infusion should only be dissolved in either 100 ml normal saline for infusion or 100 ml 5% dextrose for infusion. No other solutions for I.V. infusion should be used.
After reconstitution from a microbiological point of view, use immediately (i.e. within 3 hours) and any unused portion should be discarded. The duration of administration should be 20-30 minutes.

Use in the elderly
Dosage adjustment is not necessary.

Use in children
There is limited experience of use in children.

Impaired renal function
Dose adjustment is not required in patients with impaired renal function.

Impaired hepatic function
As half-life is increased in patients with impaired hepatic function, the dose requires adjustment and a daily dose of 10 mg-20 mg may be sufficient.

Route of Administration:
For I.V. injection or infusion after reconstitution.

Contraindication:
Known hypersensitivity to omeprazole or to any of the other constituents of the formulation.
Omeprazole like other PPIs should not be administered with atazanavir.

Warning and precautions:
When gastric ulcer is suspected, or an alarm symptom such as unintentional weight loss, vomiting, dysphagia, haematemesis occurs, the possibility of malignancy should be excluded as treatment may alleviate symptoms and delay diagnosis.

Regular Surveillance
Patients on proton pump inhibitor treatment (particularly those treated for long term) should be kept under regular surveillance.
Subacute Cutaneous Lupus Erythematosus (SCLE)
Proton pump inhibitors are associated with very infrequent cases of subacute cutaneous lupus erythematosus (SCLE). If lesions occur, especially in sun-exposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should consider stopping {product name}. SCLE after previous treatment with a proton pump inhibitor may increase the risk of SCLE with other proton pump inhibitors.
Hypomagnesaemia
Severe hypomagnesaemia has been reported in patients treated with PPI like {product name} for at least three months, and in most cases for a year. Serious manifestations of hypomagnesaemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur but they may begin insidiously and be overlooked. In most affected patients, hypomagnesaemia improved after magnesium replacement and discontinuation of the PPI.
For patients expected to be on prolonged treatment or who take PPI with digoxin or drugs that may cause hypomagnesaemia (e.g., diuretics), health care professionals should consider measuring magnesium levels before starting PPI treatment and periodically during treatment.
Fracture
Proton pump inhibitors, especially if used in high doses and over long durations (>1 year), may modestly increase the risk of hip, wrist and spine fracture, predominantly in the elderly or in presence of other recognised risk factors. Observational studies suggest that proton pump inhibitors may increase the overall risk of fracture by 10–40%. Some of this increase may be due to other risk factors. Patients at risk of osteoporosis should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.
Clostridium Difficile Diarrhea
Published observational studies suggest that PPI therapy may be associated with an increased risk of Clostridium difficile associated diarrhea, especially in hospitalized patients. This diagnosis should be considered for diarrhea that does not improve. Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.
Vitamin B12 Deficiency
Daily treatment with any acid-suppressing medications over a long period of time (e.g., longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo- or achlorhydria. Rare reports of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in the literature. This diagnosis should be considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

Interference with laboratory tests
Increased Chromogranin A (CgA) level may interfere with investigations for neuroendocrine tumours. If the patient(s) are due to have a test on Chromogranin A level, [product name] treatment should be stopped for at least 5 days before CgA measurements to avoid this interference (see section Pharmacodynamic). If CgA and gastrin levels have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of proton pump inhibitor treatment.

Interactions with other medicaments:
Due to the decreased intragastric acidity, the absorption of ketoconazole or itraconazole may be reduced during omeprazole therapy as it is during treatment with other acid secretion inhibitors.
As omeprazole is metabolised in the liver through cytochrome it can prolong the elimination of diazepam, phenytoin and warfarin. Monitoring of patients receiving phenytoin or warfarin is recommended and a reduction of phenytoin or warfarin dose may be necessary when Omeprazole is added to treatment. However, concomitant treatment with Omeprazole 20 mg orally daily, did not change the blood concentration of phenytoin in patients on continuous treatment with phenytoin. Similarly, concomitant treatment with Omeprazole 20 mg orally daily, did not change coagulation time in patients on continuous treatment with warfarin.
Plasma concentrations of omeprazole and clarithromycin are increased during concomitant oral administration. There is no interaction with metronidazole or amoxicillin. These antimicrobials are used together with omeprazole for eradication of Helicobacter pylori . There is no evidence of an interaction with phenacetin, theophylline, caffeine, propranolol, metoprolol, cyclo-Sporin, lidocaine, quinidine, estradiol, or antacids when Omeprazole is given orally. The absorption of Omeprazole given orally is not affected by alcohol or food. There is no evidence of an interaction with piroxicam, diclofenac or naproxen, this is considered useful when patients are required to continue these treatments.
Simultaneous treatment with omeprazole and digoxin in healthy subjects lead to a 10% increase in the bioavailability of digoxin as a consequence of the increased intragastric pH.
Interaction with other drugs also metabolised via the cytochrome P450 system cannot be excluded.
Co-administration of omeprazole (40 mg once daily) with atazanavir 300 mg/ritonavir 100 mg to healthy volunteers resulted in a substantial reduction in atazanavir exposure (approximately 75% decrease in AUC, Cmax, and Cmin). Increasing the atazanavir dose to 400 mg did not compensate for the impact of omeprazole on atazanavir exposure. PPIs including omeprazole should not be co-administered with atazanavir. Concomitant administration of omeprazole and tacrolimus may increase the serum levels of tacrolimus.

Pregnancy and lactation:
Well-conducted epidemiological studies indicate no adverse effects of omeprazole on pregnancy or on the health of the foetus/newborn child. Omeprazole can be used during pregnancy. Omeprazole is excreted in breast milk. Influence, if any, on the child is unknown.

Side effects:
The adverse effects reported most frequently with omeprazole have been gastrointestinal disturbances, in particular diarrhoea, constipation, abdominal pain, nausea and vomiting and flatulence. Skin rash, pruritus and urticaria have also been reported, usually resolving after discontinuation of treatment.
Central nervous system disorders reported so far include parasthesia, dizziness, lightheadedness and feeling faint, all of these usually resolved on cessation of therapy. Somnolence, insomnia and vertigo have been reported rarely. Reversible mental confusion, agitation, depression and hallucinations have occurred predominantly in severely ill patients.
Other adverse events reported rarely include arthralgia and myalgia, blurred vision, taste disturbances, peripheral oedema, hyponatremia, blood disorders including agranulocytosis, leucopenia, thrombocytopenia, pancytopenia, anaphylactic shock, malaise, fever bronchospasm, encephalopathy in patients with pre-existing severe liver disease, hepatitis with or without jaundice, rarely hepatic failure and interstitial nephritis which has resulted in acute renal failure.
Rare: Hypersensitivity reactions, Myalgia, Anaphylactic reaction, Erythema multiforma.

Subacute Cutaneous Lupus Erythematosus (SCLE)
Skin and subcutaneous tissue disorders
Frequency 'not known': Subacute cutaneous lupus erythematosus
Interstitial Nephritis
Renal and urinary disorders: Interstitial nephritis
Hypomagnesaemia
Metabolism and nutritional disorders
Frequency "not known": hypomagnesaemia.

Fracture
Musculoskeletal disorders
Frequency "uncommon": Fracture of the hip, wrist or spine.
Clostridium Difficile Diarrhea
Infections & infestations: Clostridium difficile associated diarrhea.
Fundic Gland Polyps (Benign)
Gastrointestinal disorders
Frequency "common": Fundic gland polyps (benign)
Vitamin B12 Deficiency
Metabolic/Nutritional: Vitamin B12 deficiency

Symptoms and treatment of overdose:
Over Dosage:
There is no sufficient information available on deliberate over dosage with omeprazole. However, doses up to 400 mg have not resulted in severe symptoms. Therefore, treatment of overdosage should be symptomatic and supportive. In clinical trails, intravenous doses up to 270 mg / day and up to 650 mg over a three-day period have been given without any dose related adverse effects.
Treatment:
No specific antidote for omeprazole overdosage is known. Omeprazole is extensively protein bound and is, therefore, not readily dialyzable. In the event of overdosage, treatment should be symptomatic and supportive.

Storage condition:
Un opened vial: Store below 30°C. Protect from light.
Reconstituted solution: after reconstitution stable for 4 hours.

Shelf life:
Un opened vial: 24 M from the date of manufacture.
Reconstituted solution: 4 hours at 30°C.

i) I.V. Injection
4 hours

ii) Infusion
4 hours

Incompatibilities
The stability of omeprazole sodium depends on pH, in low pH fading of the solution can occur. Solutions prepared for I.V. administration should not be refrigerated.

Instruction for use and handling
Once opened and reconstituted the product may be stored for a maximum of 4 hours at 30°C.
Any unused portion should be discarded.

Presentation:
10 mL USP type I amber glass lyo vial containing 40 mg of omeprazole.

Revised date: December 2018

Product Registration Holder and Imported by:-
UNIMED SDN BHD
53, Jalan Tembaga SD 5/2B
Bandar Sri Damansara,
52200 Kuala Lumpur.

Manufactured By:
Gland Pharma Limited
Sy. No. 143-148, 150 & 151,
Near Gandimaisamma Cross Roads,
D.P. Pally, Dundigal Post,
Dundigal - Gandimaisamma Mandal,
Medchal – Malkajgiri District,
Hyderabad - 500043, Telangana, India.

LEA-019145-01

PRODUCT NAME	Omeprazole for injection	ARTWORK No.	LEA-019145-01
COUNTRY	Unimed	SUPERSEDES No.	LEA-019145-00
CUSTOMER	Malaysia	PANTONE SHADE No's	Black
DIMENSIONS	Open Dimension: 120 x 275 ±2 mm (L x W) Folded Dimension: 60 x 35 ±2 mm (LxW)	VARNISHING/LAMINATE	NA
SUBSTRATE	Maplitho Paper	GRAMMAGE	60 GSM ± 10%,
MODE OF SUPPLY	Bundles /Tray Pack (Qty. of each such pack to be provided by the vendor)		

Form No: GPL/PP/F007-00/07-14